

AGENTOS

An AI-Powered Multi-Agent Career Planning Platform

Background

Career preparation today is fragmented: learners juggle tutorials, notes, resumes and mock interviews across disconnected tools, with no personal plan and no feedback on whether they are actually improving. Generic course platforms teach the same syllabus to everyone and never measure job-readiness.

Objectives

AGENTOS closes this gap with a personal AI career planner. A learner states a target role, a timeline and current skills; the system generates a week-by-week plan, teaches each task, scores every answer, schedules revision exactly where the learner is weak, and reports a single readiness score that converges as evidence accumulates.

System design

The backend is a multi-agent pipeline. The Orchestrator drives the flow; the Analyst grades goal difficulty; the Planner writes ordered tasks with skills and weeks; the Validator scores submissions (0-100) with a skill breakdown; the Tutor explains with examples; the Interviewer runs 5-round mock interviews; the Reporter and Coach turn data into next steps. A small trained regression model projects future readiness from progress history. All LLM calls go through one wrapper with a deterministic mock fallback, retries and repair, so the system works offline and in tests.

Key features

Week-structured AI plans with re-planning; due dates with overdue tracking; private task notes with full-text search; AI tutoring and low-score remedial tasks; Leitner-style spaced repetition reviews; timed 5-question practice quizzes; mock interviews with per-round scores; resume analysis (PDF/TXT) with skill-gap detection; streaks and achievement badges; weekly coaching summaries and career report PDFs.

Implementation and quality

Backend: FastAPI + SQLAlchemy (SQLite locally, Postgres in production), Alembic migrations, JWT authentication, per-hour LLM quotas. Frontend: React + Vite + Tailwind, no heavy libraries. Quality: 154 offline pytest tests, a 4-job GitHub Actions pipeline (SQLite, Postgres, frontend build, Docker), and hand-written migrations with full test coverage of every endpoint.

Deployment and conclusion

The application is deployed and usable today: the React frontend is hosted on Vercel and the FastAPI backend runs serverless, backed by managed Postgres. AGENTOS demonstrates that a small team can ship a production-grade, multi-agent learning system that is personal, measurable and always available.